

Forecast Errors and Electricity Price Dispersion in ERCOT

How do joint errors in short-range (HRRR 1h) and day-ahead (GFS) wind and temperature forecasts impact locational marginal prices in ERCOT?

Full year 2025 | Combined HRRR 1h + GFS day-ahead pipeline

1 Introduction

This report examines how forecast errors in two weather models — the High-Resolution Rapid Refresh (HRRR, 1-hour ahead) and the Global Forecast System (GFS, day-ahead) — affect system-level electricity price dispersion in ERCOT. The analysis uses ERA5-Land reanalysis as ground truth to compute forecast errors at each 0.1° grid cell across Texas, then links those errors to real-time locational marginal prices (LMP) at ERCOT resource nodes.

The key outcome variable throughout is **system LMP standard deviation** (`system_lmp_std`), which captures price dispersion across nodes — a measure of transmission congestion and spatial market imbalances. Forecast errors are measured for both temperature (affecting demand) and wind speed (affecting renewable generation).

All regressions absorb hour-of-day and month fixed effects unless otherwise noted. Standard errors are clustered by date except where noted.

2 Raw Correlation: Forecast Error vs. LMP Spread

Before controlling for weather conditions and time patterns, we examine the raw Pearson correlation between each pixel's forecast error and the system-wide LMP standard deviation. Each panel covers all 12 months of 2025 using all ERA5 pixels inside Texas.

Per-pixel correlation: Forecast Error vs. LMP Spread (system_imp_std)

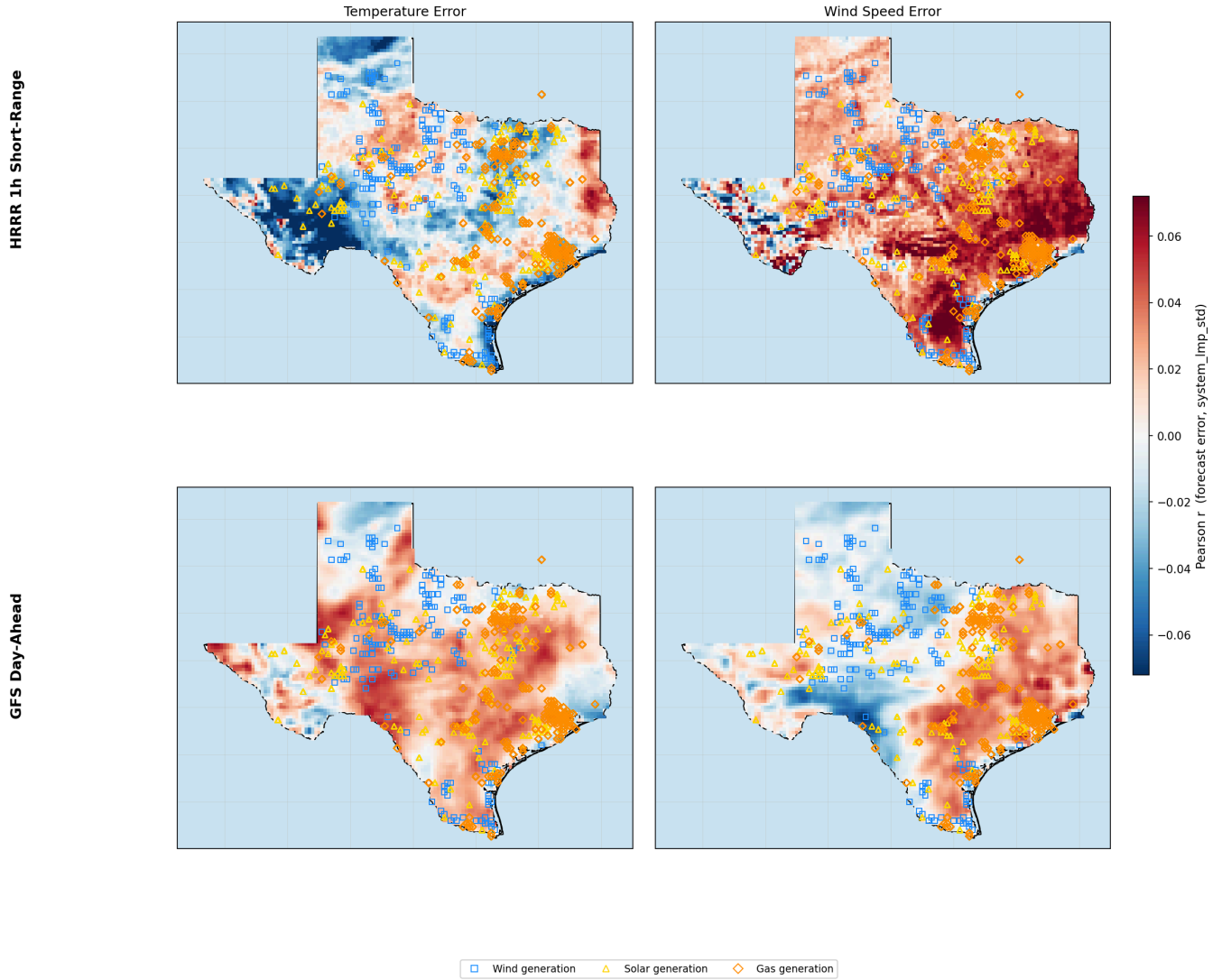


Figure 1: Per-pixel Pearson correlation between forecast error and system LMP standard deviation (system_imp_std). Top row: HRRR 1h errors; bottom row: GFS day-ahead errors. Left column: temperature; right column: wind speed. Markers show wind (squares), solar (triangles), and gas (diamonds) generation sites.

3 Pixel-Level Controlled Regression Coefficients

Each panel shows the estimated coefficient on a forecast error variable from a regression run independently for each ERA5 pixel. The regression controls for observed temperature and wind speed, a weekend dummy, and absorbs hour-of-day and month fixed effects. Only pixels with statistically significant coefficients ($p < 0.05$) are filled; insignificant pixels appear blank.

The four panels correspond to the same 2×2 layout as the correlation maps: HRRR 1h vs. GFS day-ahead $\times$ temperature vs. wind speed error.

Pixel-Level Regression Coefficients: Forecast Error $\rightarrow$ LMP Spread
(only significant pixels shown, $p < 0.05$; controls: observed weather, weekend FE; FE: hour-of-day, month)

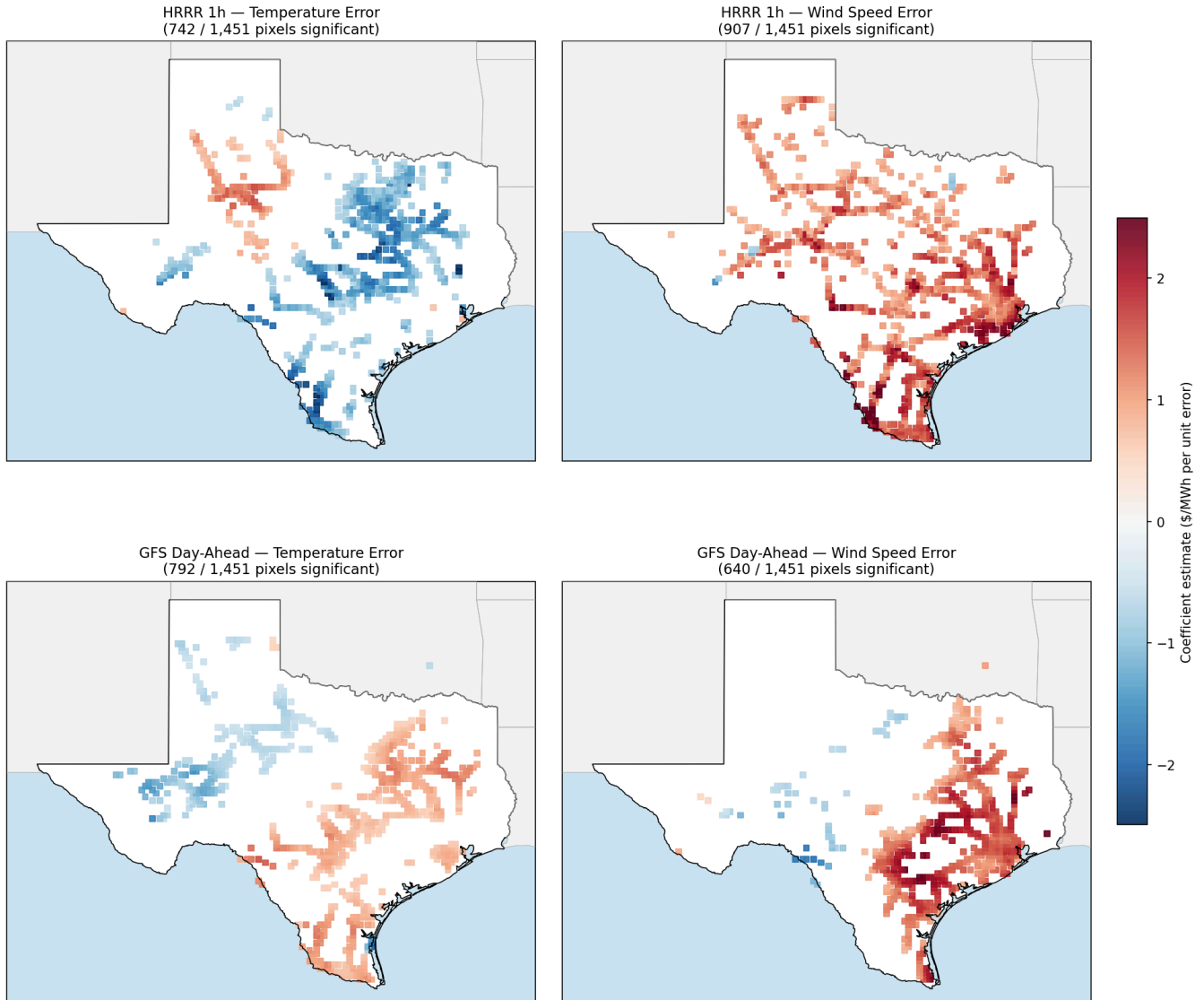


Figure 2: Pixel-level OLS coefficient estimates for each forecast error variable on system LMP standard deviation. Controls: observed ERA5 temperature and wind speed, weekend indicator. FE: hour-of-day, month. Only significant pixels ($p < 0.05$) are colored; colorbar shows \$/MWh per unit forecast error. Blank pixels are not significant.

4 Infrastructure-Level Forecast Error Effects

To understand which types of generation infrastructure mediate the relationship between forecast errors and price dispersion, we aggregate forecast errors to the infrastructure-category level. For each hour, we compute capacity-weighted mean forecast errors separately for wind, solar, gas, battery, and coal generation pixels, and unweighted means for transmission corridor and load-center pixels. This produces an hour-level dataset where each observation captures spatially-disaggregated error signals.

The regression includes cross-category interaction terms to capture amplification effects — e.g., whether temperature errors at load centers matter more when wind speed errors at wind farms are simultaneously large.

4.1 Main Regression (Full Year)

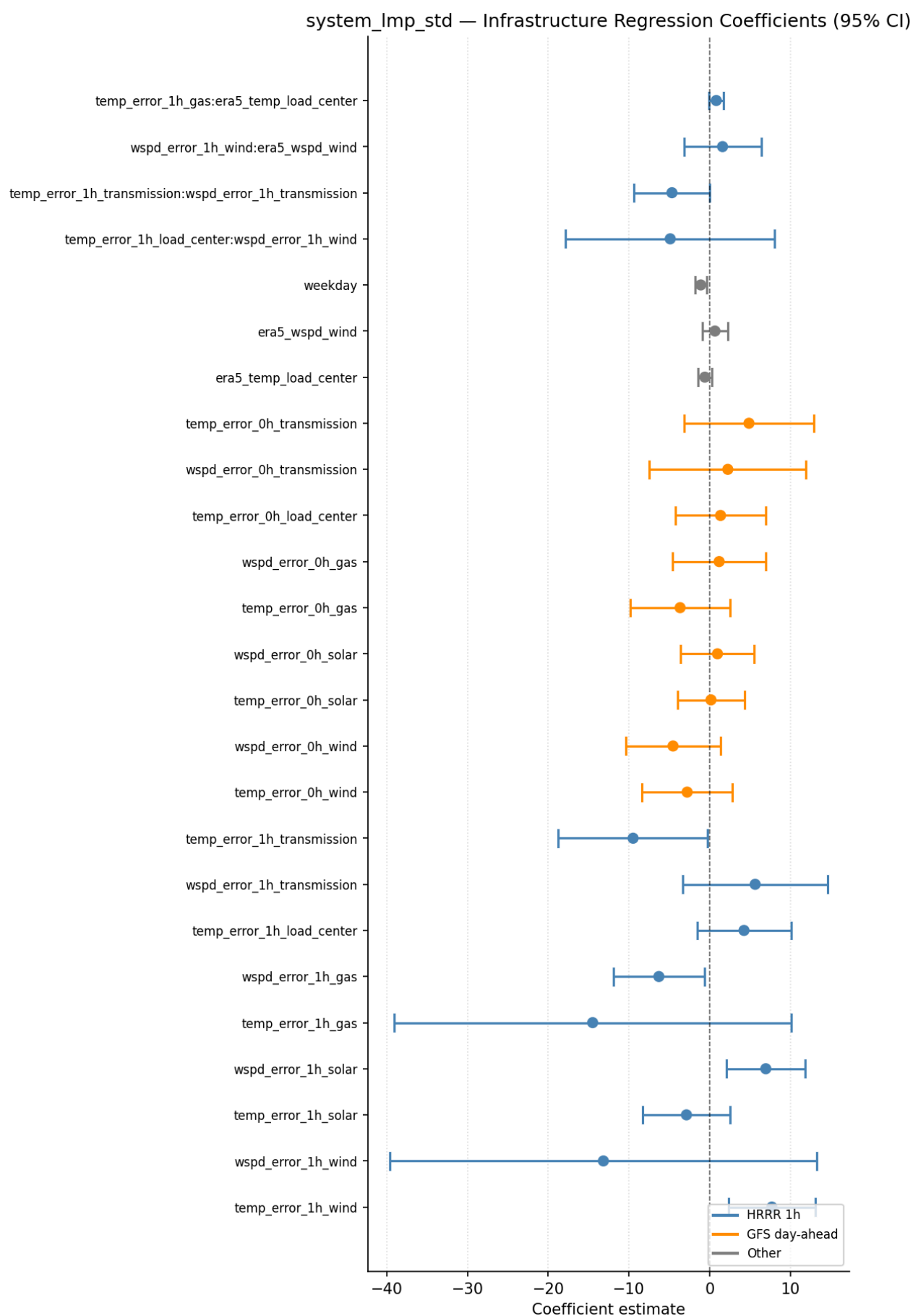


Figure 3: Coefficient estimates from the main infrastructure-level regression. HRRR 1h coefficients shown in blue; GFS day-ahead in orange. FE: hour-of-day, month. Clustered SE by date. Outcome: system LMP standard deviation (\$/MWh).

Variable	Season	Std Error	Stars	N Obs
temp_error_1h_wind	full_year	2.723	***	8398
wspd_error_1h_wind	full_year	13.489		8398
temp_error_1h_solar	full_year	2.756		8398
wspd_error_1h_solar	full_year	2.482	***	8398
temp_error_1h_gas	full_year	12.563		8398
wspd_error_1h_gas	full_year	2.872	**	8398
temp_error_1h_load_center	full_year	2.973		8398
wspd_error_1h_transmission	full_year	4.582		8398
temp_error_1h_transmission	full_year	4.723	**	8398
temp_error_0h_wind	full_year	2.861		8398
wspd_error_0h_wind	full_year	2.992		8398
temp_error_0h_solar	full_year	2.124		8398
wspd_error_0h_solar	full_year	2.330		8398
temp_error_0h_gas	full_year	3.152		8398
wspd_error_0h_gas	full_year	2.935		8398
temp_error_0h_load_center	full_year	2.861		8398
wspd_error_0h_transmission	full_year	4.950		8398
temp_error_0h_transmission	full_year	4.098		8398
era5_temp_load_center	full_year	0.444		8398
era5_wspd_wind	full_year	0.801		8398
weekday	full_year	0.357	***	8398
temp_error_1h_load_center:wspd_error_1h_wind	full_year	6.608		8398
temp_error_1h_transmission:wspd_error_1h_transmission	full_year	2.395	*	8398
wspd_error_1h_wind:era5_wspd_wind	full_year	2.442		8398
temp_error_1h_gas:era5_temp_load_center	full_year	0.469	*	8398

Table 1: Infrastructure-level regression results. Dependent variable: system LMP standard deviation. Bold coefficients are significant at the 5% level.

4.2 Seasonal Heterogeneity

system_imp_std — Seasonal Infrastructure Regression Coefficients (95% CI)

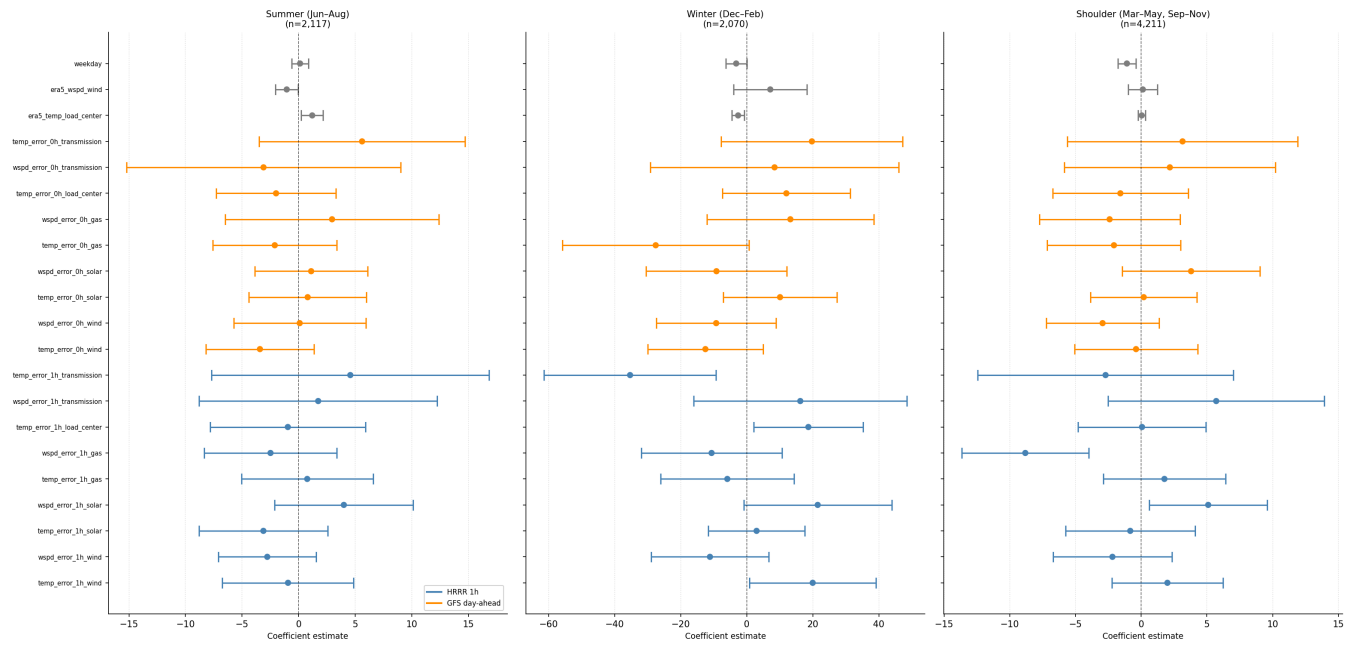


Figure 4: Seasonal subsample regressions. Left: Summer (Jun–Aug); center: Winter (Dec–Feb); right: Shoulder (Mar–May, Sep–Nov). FE: hour-of-day only. Clustered SE by date.

5 Cluster Heterogeneity

To capture geographic variation in how forecast errors propagate to prices, we cluster ERCOT resource nodes using agglomerative clustering with a geographic contiguity constraint. Nodes are clustered on standardized latitude/longitude (with geographic weight) plus LMP summary statistics. For each cluster, we run a joint regression including both HRRR 1h and GFS day-ahead error terms.

5.1 Cluster Map

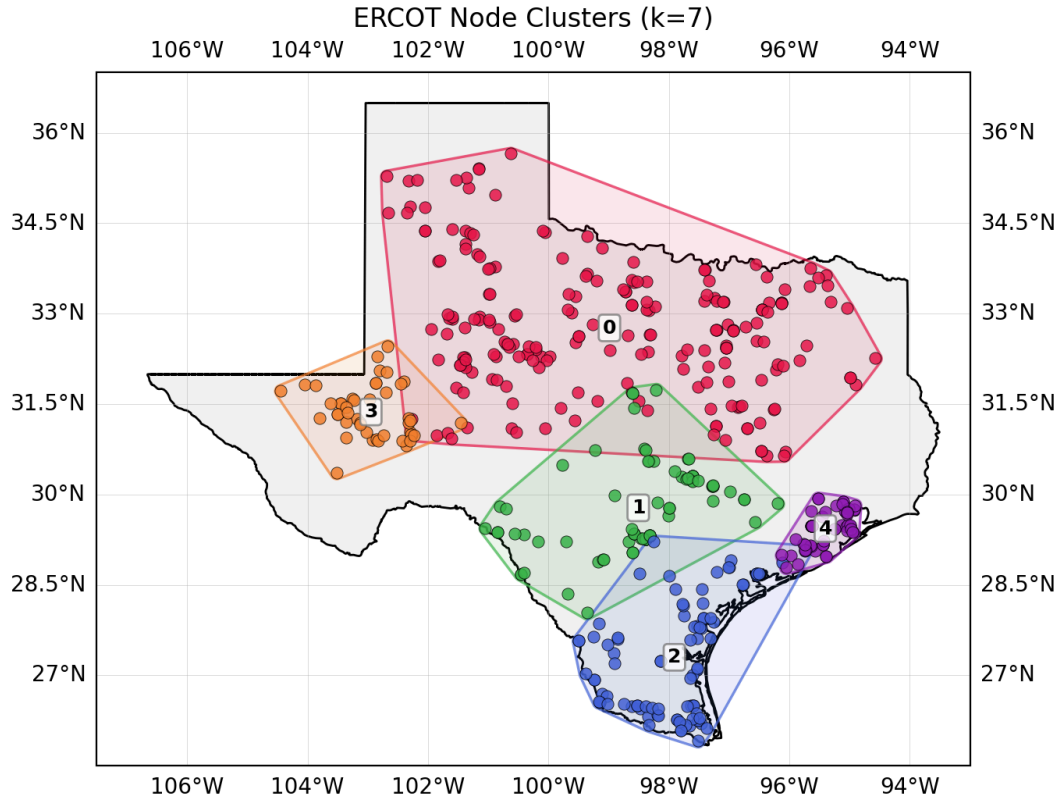


Figure 5: Geographic distribution of ERCOT resource node clusters. Formed via agglomerative clustering with geographic connectivity.

5.2 Coefficient Estimates by Cluster

Per-Cluster Coefficients — HRRR 1h (top) vs GFS Day-Ahead (bottom)

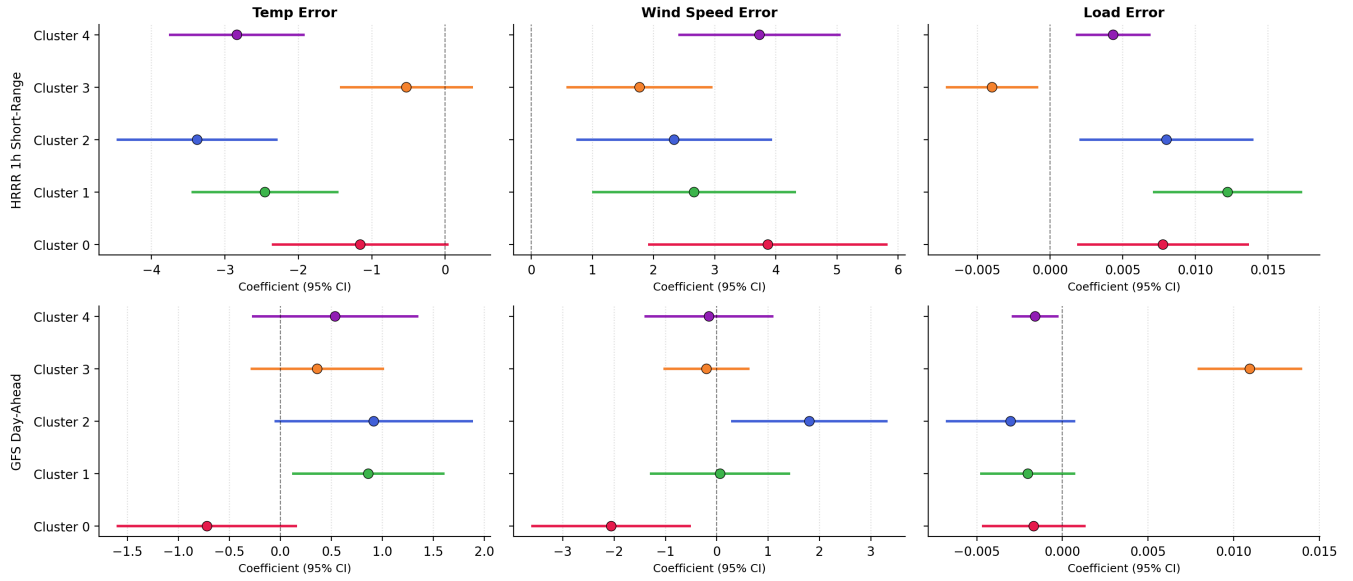


Figure 6: Per-cluster OLS estimates from a joint regression including both HRRR 1h (top row) and GFS day-ahead (bottom row) forecast errors. Error bars: 95% confidence intervals. FE: hour-of-day, month. Controls: weekday, actual load.

5.3 Distribution of Marginal Effects — HRRR 1h

Distribution of Marginal Effects by Cluster — 1h Lead
(variable $\times$ estimated coefficient)

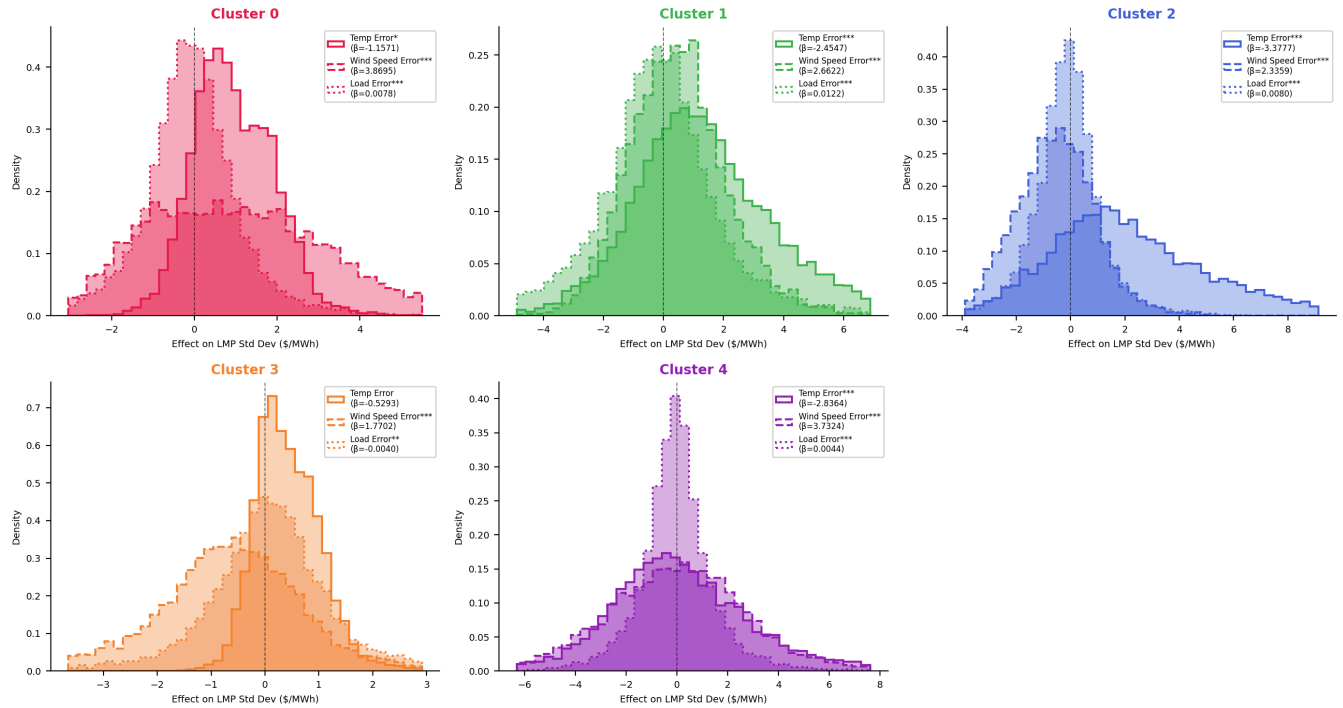


Figure 7: Marginal effect distributions for HRRR 1h errors by cluster. Each histogram shows the distribution of (variable $\times$ estimated coefficient), i.e., the contribution to LMP std dev. Solid: temperature error; dashed: wind speed error; dotted: load error.

5.4 Distribution of Marginal Effects — GFS Day-Ahead

Distribution of Marginal Effects by Cluster — 0h Lead
(variable $\times$ estimated coefficient)

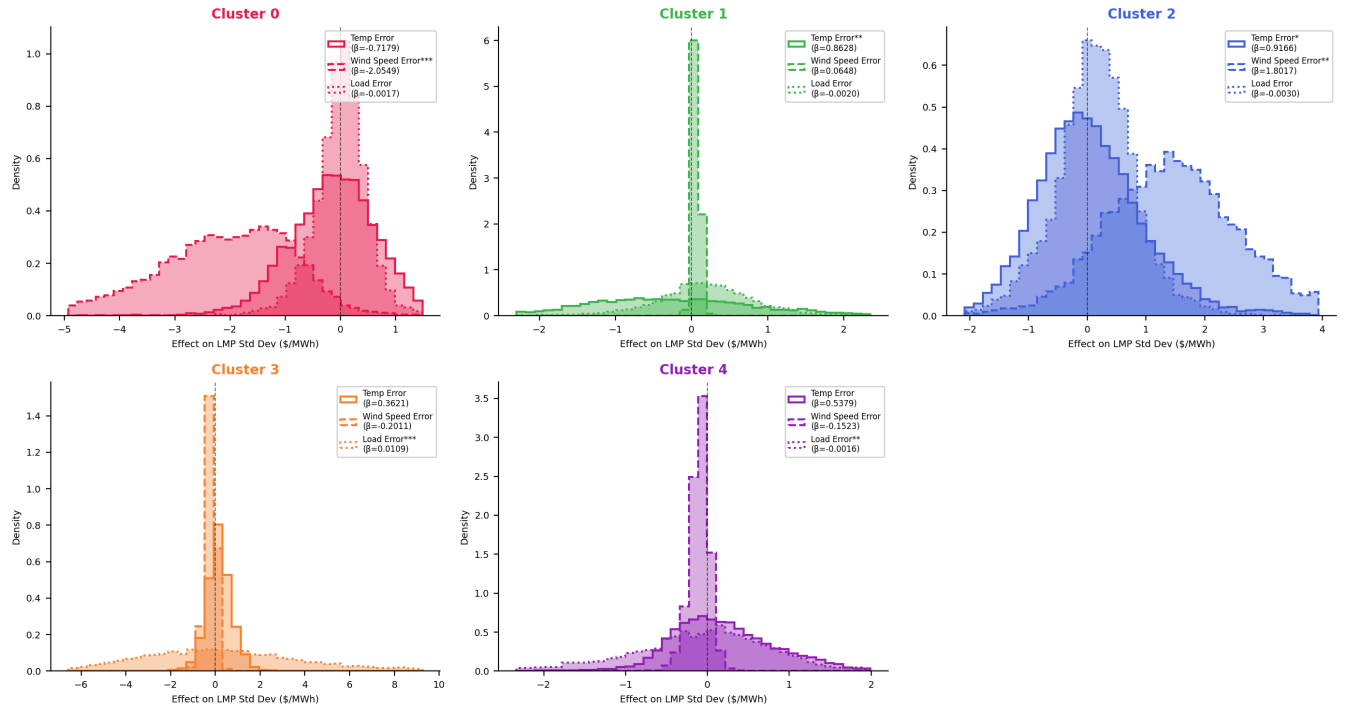


Figure 8: Same layout as previous figure but for GFS day-ahead forecast errors.

6 Graph Neural Network Results

Results pending. A graph attention network (GAT) trained on the ERCOT transmission topology will be incorporated here once training and evaluation are complete. The GNN learns to propagate nodal weather features across the transmission graph to predict locational LMP, capturing non-linear and network-mediated effects that the linear regressions above cannot recover.

7 Appendix

7.1 Cluster Regression Summary Table

Variable	Cluster	Coef	Std Error	T Stat	P Value	N Obs	Stars
temp_error_1h	0	-1.157	0.614	-1.883	0.060	7708	*
wspd_error_1h	0	3.870	0.999	3.874	0.000	7708	***
load_error_1h	0	0.008	0.003	2.578	0.010	7708	***
temp_error_0h	0	-0.718	0.451	-1.591	0.112	7708	
wspd_error_0h	0	-2.055	0.793	-2.591	0.010	7708	***
load_error_dah	0	-0.002	0.002	-1.084	0.278	7708	
weekday	0	-0.781	0.223	-3.501	0.000	7708	***
actual_load	0	0.013	0.001	23.460	0.000	7708	***
temp_error_1h	1	-2.455	0.510	-4.814	0.000	7708	***
wspd_error_1h	1	2.662	0.851	3.129	0.002	7708	***
load_error_1h	1	0.012	0.003	4.672	0.000	7708	***
temp_error_0h	1	0.863	0.381	2.263	0.024	7708	**
wspd_error_0h	1	0.065	0.697	0.093	0.926	7708	
load_error_dah	1	-0.002	0.001	-1.421	0.155	7708	
weekday	1	-0.730	0.223	-3.269	0.001	7708	***
actual_load	1	0.010	0.000	23.342	0.000	7708	***
temp_error_1h	2	-3.378	0.558	-6.048	0.000	7708	***
wspd_error_1h	2	2.336	0.817	2.860	0.004	7708	***
load_error_1h	2	0.008	0.003	2.620	0.009	7708	***
temp_error_0h	2	0.917	0.496	1.848	0.065	7708	*
wspd_error_0h	2	1.802	0.776	2.322	0.020	7708	**
load_error_dah	2	-0.003	0.002	-1.570	0.116	7708	
weekday	2	-1.006	0.227	-4.439	0.000	7708	***
actual_load	2	0.014	0.001	18.389	0.000	7708	***
temp_error_1h	3	-0.529	0.462	-1.145	0.252	7708	
wspd_error_1h	3	1.770	0.611	2.899	0.004	7708	***
load_error_1h	3	-0.004	0.002	-2.463	0.014	7708	**
temp_error_0h	3	0.362	0.334	1.085	0.278	7708	
wspd_error_0h	3	-0.201	0.429	-0.469	0.639	7708	
load_error_dah	3	0.011	0.002	7.019	0.000	7708	***
weekday	3	-1.201	0.226	-5.304	0.000	7708	***
actual_load	3	-0.019	0.002	-10.171	0.000	7708	***
temp_error_1h	4	-2.836	0.471	-6.024	0.000	7708	***
wspd_error_1h	4	3.732	0.677	5.514	0.000	7708	***
load_error_1h	4	0.004	0.001	3.317	0.001	7708	***
temp_error_0h	4	0.538	0.416	1.294	0.196	7708	
wspd_error_0h	4	-0.152	0.641	-0.238	0.812	7708	
load_error_dah	4	-0.002	0.001	-2.263	0.024	7708	**
weekday	4	-0.891	0.230	-3.880	0.000	7708	***
actual_load	4	0.006	0.000	17.101	0.000	7708	***

Table 2: Per-cluster regression coefficients. Joint HRRR 1h + GFS day-ahead specification. Bold: significant at 5%.